UNIVERSITY OF MAURITIUS

FACULTY OF INFORMATION COMMUNICATION & DIGITAL TECHNOLOGIES

Synopsis

"How can Virtual Reality raise awareness about climate change and its impacts?"

Module: Communication & Business Skills for IT

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1 Introduction

1.1 Background and Problem Rationale

Climate change remains the defining global challenge of the twenty-first century, with its effects becoming increasingly visible and severe across planet. The Intergovernmental Panel on Climate Change (IPCC,2023) reports that global temperatures have already risen by approximately 1.1ºC above pre-industrial levels, triggering severe consequences. The energy sector remains the primary contributor to this crisis, accounting for approximately 73% of global greenhouse gas emissions (Our World in Data, 2024). However, despite decades of scientific consensus and awareness campaigns, global emissions continue to rise.

This persistent gap between knowledge and behavioural change seemingly stems from a dire communication failure. Traditional methods of communicating climate information-such as scientific reports, static data visualizations, and documentary media- rely on passive information delivery. Research indicates that abstract statistics and technical terminology often fail to create the emotional engagement necessary for individuals to connect with climate issues on a personal level (Wang et al., 2022). This can be attributed to the fact that complex technological terminologies and abstract statistics often present climate change as a psychologically distant threat. Since the audience cannot personally connect with the problem, facts alone are insufficient.

The Workplace Challenge

This disconnect is particularly emphasized in organizational and workplace settings, which emerge as a critical yet underexplored context for energy conservation. In these environments, energy use is consequently high, but reducing it proves to be difficult as established company cultures and shared responsibilities shadow individual accountability. Although employees may have a broad understanding of climate issues, the abstract nature of climate change leaves them uncertain of how their daily workplace actions matter or how to implement changes that do matter.

Virtual Reality as a Catalyst for Change

To address these limitations, an interactive and emotionally resounding communications tool is clearly needed. Virtual Reality(VR) has evolved significantly in recent years, surfacing as a powerful solution by generating a strong feeling of presence. By simulating immersive environmental experiences, such as rising sea levels or melting ice caps, VR bridges the gap between intellectual understanding and emotional connection. Research shows that experiencing environmental degradation in VR increases empathy and motivates individuals to adopt sustainable behaviours.

This research proposes the development of a VR-based climate visualization and energy conservation system. By combining immersive environmental simulations with interactive decision-making scenarios, the proposed system aims to transform how individuals understand their relationship with energy consumption and climate impact, ultimately instilling pro-environmental habits within them.

1.2 Aim, Objectives and Scope

AIM:

To propose a conceptual framework for a VR-based climate visualization and energy conservation system that enhances public understanding of climate change impacts and motivates sustainable energy behaviours.

OBJECTIVES:

- To **analyse** the limitations of traditional data communication methods in fostering engagement with sustainability goals.
- To **evaluate** the psychological mechanisms, specifically spatial presence and empathy, through which VR environments influence the environmental behaviour of an individual.
- To **conceptualize** a VR system that translates abstract energy metrics into immediate, observable environmental impacts such as flooding simulations.
- To **establish** a framework for measuring the impact of this VR system on daily energy conservation practices.

SCOPE

INCLUSIONS:

1. **Conceptual Framework:** Development of a high-level system design, including proposed features, user interactions, and technical requirements for the VR experience.
2. **VR applications:** Examination of existing VR environmental projects to identify best practices and design considerations.
3. **Behavioural Focus:** Exploration of how immersive experiences can influence attitudes toward energy conservation and climate action.
4. **Target Audience:** Primary focus on public users and workplace/organizational settings where energy consumption patterns can be addressed.

EXCLUSIONS:

1. **Hardware Development:** The study will not involve the design or manufacturing of VR hardware; it assumes the use of existing commercial VR devices.
2. **System Deployment:** This research does not include the programming, coding, or full implementation of a functional VR application. It remains at the conceptual and proposal stage.
3. **Macro-Industrial Analysis:** Since the focus remain on individual behaviour, heavy industrial, agricultural and national grid energy policies are omitted.

2 Background Study

2.1 Overview of Climate Change and Global Energy Trend

Climate change, driven primarily by anthropogenic greenhouse gas emissions, has led to global surface temperatures rising by approximately 1.1ºC above pre-industrial levels- with each successive decade warmer than any preceding since 1850. This warming triggers severe consequences, such as extreme weather conditions, rapidly melting ice caps and significant biodiversity loss. Coastal and island regions, such as the Maldives, face existential threats, from the rising sea level and coastal flooding, on territorial integrity and economic stability. Furthermore, these shifts have severe socio-economic impacts, exacerbating food insecurity, water scarcity, health crises and widespread infrastructure damage.

The primary catalyst for this crisis remains our deep global dependency on fossil fuels. A 2016 study from 29 European countries found out that fossil fuel dependency remained high, with more than 60% of total energy coming from fossil fuel consumption among 24 of these countries. Although renewable energy is gaining popularity, with global solar capacity exceeding 2 TW and wind power reaching 1,245 GW in 2025, fossil fuels continue to dominate electricity generation and industrial activities. Consequently, global CO2 emissions from fossil fuels will hit a projected record of 42.2 billion tonnes in 2025. Unfortunately, the high costs of infrastructure, economic pressures on businesses and a rapidly growing global energy demand hinder the transition to low-carbon technologies.

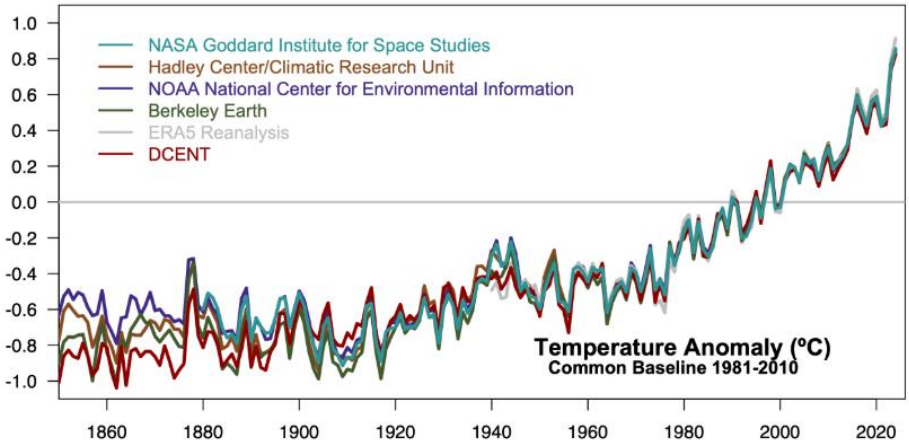
2.2 The Communication Gap and the Role of Virtual Reality

Despite treaties like the Paris Agreement, public engagement regarding climate action remains stagnant du to the limitations of traditional climate communications. Standard communication relies heavily on the “Information Deficit Model”. This model wrongly assumes that presenting more scientific facts automatically leads to behavioural change. However, static data and abstract reports often frame climate change as a psychologically distant thread, birthing “Climate Perception Inertia” rather than emotional engagement.

Virtual Reality(VR) helps to bridge this gap by serving as a transformative medium that replaces passive information consumptions with active, experiential learning. By allowing a sense of ‘presence’ and allowing users to virtually experience climate impacts, VR helps reduce the psychological distance and instils the empathy to promote impactful and pro-environmental behaviours.

2.3 Limitations of Traditional Climate Communication

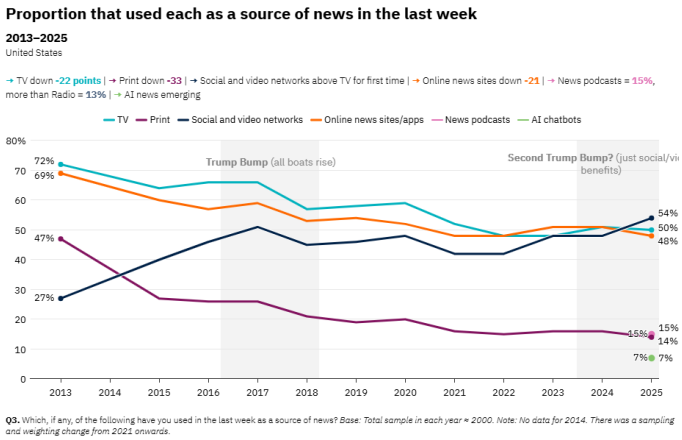
2.3.1 Static Data Representation



With the Paris Agreement treaty adopted in 2015 by 196 parties, where an ambitious goal of keeping the global temperature rise since the pre-industrial era below 1.5 Degrees Celsius was set, after nearly a decade of scientific consensus and global communication campaigns, individual action of climate change stills seems to be stagnant. Reserved to a relatively select audience, traditional climate communication, primarily reliant on scientific reports, abstract data and fear-based narratives, is therefore not be sufficient to address this pressing issue of climate change.

It is often argued that the lack of emotional connection is the primary barrier to behavioural. Virtual Reality can serve as a transformative medium, capable of replacing the traditional text-and-image-based communication by an experience that can foster presence, albeit virtually, and instil empathy, ultimately reducing psychological distance and promoting energy conservation. VR shifts the focus from passive consumption of information to active and experiential learning, hence aiding the necessary motivation to move audiences from apathy to action.

2.3.2 Public Engagement Challenges



For decades now, we have adopted the "Information Deficit Model" as a standard for communication. It assumes that when presented with more facts, people are more likely to act. However, empirical data says otherwise. Contrarily to its supposed impact, traditional media rather creates a sense of hypothetically and temporal distance. With 2024 and 2025 being the hottest years ever recorded, public engagement seems to remain trapped in a state of "Climate Perception Inertia".

The Psychological Distance Framework

Climate change is often viewed as a problem that is far away in time or space and hence, making it irrelevant to daily life. Psychological distance remains the most persistent barrier to energy conservation. Even as global temperatures exceed the 1.5 degrees Celsius threshold for 21 out f 24 months leading into July 2025, many still characterize climate change as a spatial and temporal abstraction. According to the Construal Level Theory, people perceive events as distant across four dimensions:

- Temporal: It will happen in 205, not now.
- Spatial: It affects the Arctic, not my backyard
- Social: It affects developing nations, not my community
- Hypothetical: The projections are uncertain.

Moreover, we can cite the Action Paradox, where of the 67% of people who report being concerned by climate change, only 25% believe that their individual actions make a measurable difference.

Climate Fatigue

Research from the Reuters Institute in 2025 indicated that news consumption regarding climate change in major economies, like the UK, USA and France, has declined, driven by the desire to avoid the helplessness associated with the topic.

Climate fatigue, or compassion fade, occurs when continuous exposure to dire climate news often leads to emotional exhaustion rather than action. For example, if someone was shown the imagery of impending apocalypse, that person's defensive psychological mechanism would be triggered, leading to disengagement rather than empowerment.

As per the climate change and news audiences report 2025, despite the increasing frequency of extreme weather reporting, public interest in climate news seems to have either fallen or remained still in 5 out of 8 major countries.

For reference:

Country	News Interest Change (2022-2025)	Trust in media for Climate Info
Usa	-11%	43%
UK	-7%	44%
France	- 5%	36%
India	+3%	62%
Pakistan	+1%	72%
Global Average	-2%	49%

Information Overload

Pollution, climate change and biodiversity loss can be referred to as the Triple Planetary Crisis and information regarding these is often relayed through a 'firehose' of data that exceeds human cognitive limits. This consequently leads to Information Overload, where the brain defaults to simplified narratives or unfortunately complete disengagement.

Furthermore, information disorders, simply misinformation, tend to further suppress public engagement. According to Reuters, on average, 25% of the global population reports seeing false or misleading climate information on a weekly basis, creating a distrust loop.

This loop can be summarized in 3 words: Overload, Confusion and Withdrawal.

"The human response to the climate crisis is being obstructed not by a lack of data, but by the circulation of misleading narratives that exploit the complexity of the science." - IPIE Systematic Review, 2025.

2.4 Virtual Reality in Education and awareness

2.4.1 Overview of Virtual Reality Technology

Definition of VR: Virtual Reality (VR) is a fully digital, computer-generated, three-dimensional experiential environment. Within virtual reality, it is possible for a human to physically handle virtual things or animated objects. Users' senses are heightened and active just as they are in the real world. This is useful to raise awareness in users by making them experience scenarios that are predicted to occur in the future or that are too dangerous.

Key components: A Virtual Reality system is an integrated system that consisting of both hardware and software component for the smooth and accurate running of the system. The components are:

- **Virtual Reality Headset:** The Virtual reality headset is the backbone of the VR system. It utilizes near-eye display systems to deliver immersive and mixed reality experiences. It also includes:

- **Controllers:** Virtual reality motion controllers are typically considered to be gadgets that are held in both hands. They include depth-sensing cameras, accelerometers, and gyroscopes. As Ming(2022) states, these components monitor the position of the user's head and offer the system an additional sensory input with which it can interact. Controllers allow users to perform the following actions: pointing, gesturing and grabbing.
- **Rendering system:** Another important part of Virtual reality is its rendering system. The latter is crucial to maintain immersion and prevent user discomfort (like motion sickness). It allows VR systems to translate complex scenes at a minimum of 90 frames per second. It makes use of powerful Graphics Processing Units to fully transform offline, pre-rendered imagery to real-time visuals.

2.4.2 Applications in Environmental Education

1. "Virtual Reality Climate Change" Project

This is a project developed by the United Nations. It uses VR to take users on a journey through a series of climate change scenarios, such as rising sea levels, extreme weather events, and the loss of biodiversity. By experiencing these scenarios firsthand, users gain a deeper understanding of the global implications of climate change and the need for immediate actions.

2. "Virtual Reality Amazon" Project

This project allows users to explore the Amazon rainforest in VR, providing a firsthand look of the biodiversity of this critical ecosystem. By immersing users in the sights and sounds of the rainforest, the project helps raise awareness about the threats facing the Amazon, such as deforestation and illegal logging. Hence, VR is being used to explore and protect endangered habitats.

3. "Rewilding VR" Project

In addition to exploring existing habitats, VR can also be used to simulate the restoration of degraded ecosystems. The "Rewilding VR" Project, for example, allows users to participate in the restoration of a degraded landscape, planting virtual trees and reintroducing wildlife to a barren environment. This immersive experience not only educates users about the importance of habitat restoration but also empowers them to act in the real world.

4. In schools

Despite its significant costs, VR in schools has proven to be very beneficial for the students. It actively engages students, fostering a deep understanding of complex eco-sustaining motivation and developing critical awareness. This transformative process leads to greater environmental awareness and inspires attitudes and actions that support ecosystem conservation. Focusing specifically on plastic and water pollution, VR experiences have been shown to deepen students' understanding of the environmental impact of pollution. The students expressed a strong desire to adopt eco-friendly habits such as reducing plastic use and recycling. This highlights the potential of VR to not only raise awareness, but also inspire real, meaningful actions. (Ying et al., 2024)

5. In museums

The use of VR in museums has been a real gamechanger. It has the potential to fully immerse visitors in a simulated environment where extinct animals are still roaming planet Earth. This gives them the opportunity to see the animals up close, monitor their behaviours and unique characteristics and even interact with them. The visitors are also made to experience the extreme environmental conditions that pushed the animals to go extinct. Consequently, this will encourage them to adopt eco-friendly practices as they will be keen to protect other remaining species.

6. Simulation-based learning

Simulation-based learning is a more prominent and practical approach to sustainable development compared to theoretical knowledge. Simulations can be built in a way so that students become corporate managers in a virtual environment and they must make various decisions to sustainably develop the simulated environment. The students will therefore engage with technical knowledge such as natural resource management, energy efficiency and environmental

policies. This will sharpen their problem-solving and critical thinking skills and aid them in identifying the factors that aim at reducing climate change effects. The students will be encouraged to put these measures into practice even if it is on a smaller scale.

2.4.3 Effectiveness and Limitations of VR for Behavioural Change

VR vs. Traditional Media

Traditional communication often traps audiences in an "Abstract Saturation" and "Fear Paralysis".

- ***Abstract Saturation*:** When data is presented as CO2 parts per million or global mean temperature increases of 1.5 Degrees Celsius, our brain struggles to attach an emotional value to these numbers.
- ***Fear Paralysis*:** Constant exposure to extreme headlines without a personal connection often leads to emotional fatigue and our subconsciousness disengages as a defence mechanism to avoid anxiety.

VR counters this by situating the user in the psychological state of being 'present', effectively closing the feedback loop between behaviour and consequence.

For example, when a person "experiences" the heat stress of a drought or walks through a forest following a wildfire via VR, it becomes a lived memory. It shifts the user's mindset from "I know this is happening" to "I feel this and therefore, I must act."

The effectiveness of either means of communication compare as such:

Metric	Traditional Media(Text/2D)	Immersive VR	Source
Self-Efficacy Gain	Around 12%	Around 70%	Emerald Publishing (2026)
Indifference Level	High	Significantly Reduced	Stanford Report (2025)
Awareness Retention	Low	High	Scientific Reports (2025)
Risk Perception	Moderate	High	MDPI Education (2025)

The most significant impact of VR is its ability to collapse psychological distance. A 2025 Stanford-led study published in Scientific Reports used VR to transport users to distant, climate-impacted communities.

The key findings were as such:

- The participants were considerably less dismissive of climate impacts occurring in faraway locations.
- Unlike traditional media, VR instilled a sense of "Constructive Frustration" which is an emotional state that encourages problem-solving rather than paralysis.
- Merely one week post the experience, it was found that users' intentions to reduce energy or plastic consumption remained measurable.

The Limitations of VR

However, while VR has proven its effectiveness, its ability to foster permanent behavioural change is challenged by systematic limitations.

One significant barrier is the Cognitive Load. While 'Flash Points' represent sudden, catastrophic events like coastal floods and wildfires, the 'Boiling Frog' effect illustrates slow, incremental environmental degradation. VR struggles to close the gap between a one-time peak immersive experience and the boring, repetitive nature of energy conservation, such as turning off the lights daily.

Furthermore, phenomena like "Empathy Decay" show that without real-world reinforcement, pro-environmental actions tend to revert to baseline levels within weeks of the VR simulation.

3 Proposed Virtual Reality Experience

3.1 System Concept Overview

The system will consist of a VR headset, controllers and a fully functional rendering system. It will be built based on making the harsh effects of climate change known to people so that they are able to reflect on their actions. The system will simulate extreme weather events such as floods, storms and wildfires and the modelling of the gradual evolution of rising sea levels and environmental degradation. Rather than relying on static, pre-programmed environmental responses, the system will integrate a probabilistic code to dynamically generate environmental feedback, and it will also process user choices concerning energy consumption and instantly derive and display the impact that they have on the environment.

Our target users will be:

1. **Students, the future of tomorrow -** It is extremely important that we educate the society about the consequences of climate change as from a very young age itself. The VR system will aid the students to develop an ecofriendly mindset.

2. **The public -** We need the public to make an impact. Our goal of widespread environmental awareness can only be achieved if everyone in the society decides to change their bad habits and make environment conservation a priority.

3. **Policymakers –** They will experience the effects of climate change firsthand and act as a gateway to a broader audience. They will sense an urgency to act on the matter of climate change, and this may lead to sensitisation campaigns and the implementation of strict laws for the protection of the environment.

Core features:

1. The system gives the user the ability to be completely **immersed** in the virtual environment by blocking distractions from the physical world with the use of high-definition visuals and high-quality sounds.
2. The VR system puts the user in a state where he feels that he is physically **present** of the simulated world. It makes it easier for the user to have real emotions about the visuals.
3. The user is not only a passive observer, instead he can **interact** with his surroundings and input his choices (e.g. concerning energy consumption). This allows users to get the whole experience and feel part of the virtual environment.
4. The VR system needs to be powerful enough to provide **real-time response**. This is crucial to maintain user's attention in the virtual reality environment. Responding instantly to user inputs or movements will sustain immersion.
5. The VR system will simulate **three dimensions visualization** to create more depth and for it to look more realistic. This also helps users to better understand complex environments, such as landslides due to a flood.

3.2 Visualization of Climate Change Impacts

VR experiences leverage high-fidelity visualization of climate impacts allowing users to move beyond abstract sensitization and to witness the hyper-local consequences of a warming planet. By simulating extreme weather conditions, alongside gradual environmental degradation, VR shifts the narrative from just "global statistics" to "personal survival".

As aforementioned, VR creates the "Boiling Frog" vs "Flash Point" Paradox. Nonetheless, it still displays far better results than traditional media on both counts.

3.2.1 Simulation of "Flash Point" Events

These are sudden, catastrophic events. If a user in the simulation consistently chooses high-energy-consumption behaviours (e.g., leaving HVAC systems running in empty virtual rooms), the probabilistic backend may trigger a rapid localized disaster, such as a severe coastal flood breaching the virtual office windows. This induces immediate emotional stakes.

The impact of simulation for "Flash Point" events can be summarized as follows:

Disaster Type	VR Application Example	Core Metric	Outcome
Floods	Feel Safe VR (Italy)	Evacuation Speed	40% Faster Task Completion
Wildfires	Wildfire-Response VR (University of South Australia)	Agency/Self-Efficacy	70% Increase in Personal Agency
Storm Surges	Weather the Storm(NOAA)	Emotional Engagement	Affective Heuristic Triggered

3.2.2 Simulation of "Boiling Frog" Events

These illustrate slow, incremental environmental degradation. As Ying et al. (2024) note, this highlights the potential of immersive environments to accelerate cognitive recognition of long-term threats. Users can manipulate a temporal slider within the VR dashboard to watch how minor daily energy waste compounds over decades, resulting in drought or resource scarcity outside their virtual office.

Conveniently, VR can be used to demonstrate the more slow-paced issues of climate change, such as sea-level rise and environmental degradation.

<u>Visualizing Sea-Level Rise</u>

Rising sea levels are regarded as a slow-motion disaster, making them relatively easy to ignore. VR addresses this by using Bathtub Modelling alongside Hydrodynamic Modelling which conjointly factor in static elevation rise, waves, tides and bathymetry.

Modern VR experiences, such as the Rise Up project in California, integrate data from the USGS Coastal Storm Modelling System (CoSMoS). As per a study carried out for the Stanford Report in 2025 involving 163 participants, it showed that those who virtually visited rising sea-level via VR were significantly less dismissive and reported higher frustration levels, which is a key motivator for engagement.

<u>Visualizing Environmental Degradation</u>

Time-Based Projection Scenarios is one of the most powerful tools in climate sensitization. Conveniently, VR allows for the "compression of time", allowing a user to witness decades of environmental decay in minutes.

In 2025, the University of Dresden used VR to confront users with Before and After Scenarios of a glacier melting and deforestation. The conclusion was that environmental awareness, measured via the New Ecological Paradigm scale, was significantly higher after experiencing the simulated degradation.

Consequently, the above can be summarized as such:

Scenario Type	Visualization Method	Psychological Outcome	Long Term Impact
Sea-level Rise	Data-Driven CoSMoS modelling	High Risk Perception	Increased Policy Support
Glacier Melting	Before/After Time Compression	High Awe/Grief	Heightened Awareness
Deforestation	360° Iterative Degradation	Loss of place attachment	Direct behavioural Intent

Another study carried out by Frontiers in 2025 demonstrated that embodied time travel, where a user sees themselves as a child in the past, an adult in the present and an elderly person in a degraded future, allowed participants to exhibit maintained behavioural changes like carbon footprint reduction for over 6 weeks after the experiment.

Additionally, participants in a 2026 VR simulation, carried out by ResearchGate, regarding environmental plastic degradation showed a considerable reduction in actual plastic consumption 1 week after the experience, irrespective of their beliefs pertaining to climate change.

3.3 Energy Conservation Module

3.3.1 Interactive Decision- Making Scenarios

1. User choices (renewables vs fossil fuels)

The choice between renewable energy and fossil fuels is influenced by economic and social factors. Studies in Poland and Thailand show that even though there has been a widespread use of fossil fuels in the past, people are now choosing to use renewable energy sources to decrease environmental damage.

People who are concerned about pollution and climate change are more likely to make use of renewable energy sources, especially when the technology is easy to understand and use.

A VR experience can help users visualise the impact of their energy choices, by showing the environmental effects of fossil fuels. Through immersive scenarios, users can see how switching to cleaner energy sources can help reduce carbon emissions.

2. Household energy decisions

Many households still rely on fossil fuels because they are easy to get and often cheaper. However, more households are starting to switch to renewable energy as people become more aware of environmental issues.

VR simulation could allow users to explore a virtual household and make different energy choices such as the use of solar panels or continuing to rely on non-renewable energy sources. The experience would also show how environmental awareness and social influence affect these decisions taken.

It would also highlight difficulties such as high installation costs or household debts that discourage the adoption of clean energy sources. VR can therefore help users understand how their household choices affect energy use and the environment through and interactive simulation.

3. Transportation options

Transport choices have a big influence on carbon emissions and energy consumption Sustainable choices like public transport, biking, and electric vehicles help reduce greenhouse gas emissions and improve air quality.

Replacing short flights with trains reduce emissions up to 40 times, while high-speed electric trains can lower emissions by 75 to 95% as compared to road or air travel.

Social and individual factors matter too: women often choose greener transport, while in some European countries men use 70–80% more energy, showing how habits and gender differences shape transport use.

Through the VR experience, users could compare different transport options and see their environmental impact and help raise awareness and encourage more sustainable choices.

3.3.2 Carbon Footprint Visualization

1. Real-time emission calculations

Real-time emission calculations can be incorporated into VR simulations to show users the environmental impact of their actions. As users make decisions, the system automatically calculates and displays the amount of CO_2 produced. Simulations like "Melting Sea Ice" provide users with specific tools to monitor these calculations, including CO2 concentration meters, ozone (O3) concentration meters, and sea level meters. Research shows that immersive virtual environments help better understand the consequences of their decisions and help them learn about sustainable energy options.

2. Visual indicators (smog, temperature rise)

Visual indicators can be used in VR simulations to show the environmental consequences of user choices. When users select high-emission options, the virtual environment shows the environmental consequences such polluted skies or rising temperatures, whereas choosing cleaner options results in healthier environmental conditions. Certain mobile AR platforms, like 'Aire AR,' can show real-time air quality data, making invisible pollutants such as smog visible to users. This helps users see the direct effect of their choices on the environment.

Studies made on immersive VR for climate change awareness shows that visualising environmental consequences can encourage people to adopt sustainable habits.

3.3.3 Real-Time Feedback Mechanisms

This system is essential to the effectiveness of the Energy Module Conservation Module. It allows real-time feedback to link decisions taken by the user to measurable environmental outcomes.

1. Score System

The system incorporates dynamic updates of scores based on the user's energy choices. Positive actions such as the adoption of renewable energy sources, reduced electricity use, and the minimisation of transport emissions is positive to the score while fossil fuel dependency or the excessive energy usage categorised as negative actions decrease the score. The energy sector is responsible for approximately 73% of global greenhouse gas emissions (IEA, 2023), making the energy behaviour central to climate crisis.

2. Environmental Impact Display

The module allows the user to visualise environmental consequences triggered by user decisions. CO2 concentration levels, global temperature rise and sea-level simulations are included in this projection system. The global surface temperature has expanded by around 1.1 °C above non-industrial metrics (IPCC, 2023). Research indicates that an immersive visualisation reduces psychological distance and strengthens climate risk perception. (Meijers et al., 2023).

3. Behavioural Enforcement

This system is implemented through adaptive environmental feedback. Sustainable decisions ensue in ecological restoration and more stabilised climate metrics, while unsustainable actions heighten climate impacts. Immersive VR Exposure has been shown to crucially strengthen green consciousness (Thoma et al., 2023).

4 Methodology

4.1 Research Approach

This immersive study adopts a Design-based research (DBR) approach, integrating technological progress with the theory of environmental education and the study of sciences. The project is a combination of qualitative peer-reviewed academic literature and system design methodology to develop a VR-based climate education framework to evaluate behavioural effectiveness.

Design-based research is the most convenient as the project involves iterative conceptual development of an immersive study intervention supported in empirical findings on VR and climate communication (Pellas et al., 2022).

4.2 Data Collection Sources

Data and theoretical foundations were extracted from peer-reviewed academic journals, institutional climate reports, and empirical studies on behavioral psychology. Key institutional sources include the Intergovernmental Panel on Climate Change (IPCC) and the World Economic Forum, which provided the foundational metrics regarding global emission targets and socio-economic risks. Furthermore, psychological data regarding empathy retention, cognitive load, and the "Information Deficit Model" were collected from leading environmental psychology publications to inform the behavioral architecture of the VR simulations.

4.3 Analysis of Existing VR and Climate studies

Thoma et al. (2023) conducted a controlled study with 142 participants comparing immersive VR with non-VR metrics. Findings revealed that there was a significant expansion in green awareness in VR groups and higher presence ratings in VR compared to Conditions projected. This study also resulted in a limited but positive trends in pro-environmental behaviour intention. As VR technology becomes more accessible, integrating the system into educational systems, public outreach, and policy communication may represent a crucial improvement in global climate literacy and eco-friendly behaviour.

In a similar experience, Meijers et al. (2023) noticed the increase in perceived climate risk and emotional engagement resulted from VR exposure.

Strengths identified:

- Heightened emotional connection to climate crisis.
- Reduced psychological distance.
- Improved engagement and retention
- Stronger experiential learning outcomes.

Gaps identified:

- High development and hardware costs.
- Limited long-term behavioural measurement.
- Small sample size in existing studies.

4.4 Expected Outcomes

Given that the scope of this project is strictly confined to only the theoretical research and conceptualization, the expected outcomes focus on establishing a rigorous academic foundation rather than the actual software development. The deliverables of this study will include:

A conceptual VR Framework: A detailed theoretical architecture mapping how workplace energy metrics can be translated into immersive, probabilistic environmental simulations.

Psychological Impact Synthesis: A comprehensive evaluation of the cognitive mechanisms, specifically spatial presence and empathy, required to overcome “Climate Perception Inertia” and motivate pro-environmental behaviours through the integration of real-time environmental data. It will also help foster a stronger comprehension of energy impacts by linking personal energy decisions to measurable carbon emissions.

Proposed Evaluation Method: A structured research design detailing how future studies can empirically test the VR framework’s effectiveness on the long-term behavioural change, including recommended metrics for tracking empathy and actual energy conservation.

Ultimately, this research will serve as a blueprint, bridging the gap between human-computer interactions and sustainability initiatives, laying the foundation for future technical deployment.

5 Challenges and Limitations

While the proposed framework offers a great approach to climate communication, some inherent challenges and limitations must be acknowledged within the scope of this research.

5.1 Methodological Limitations

Since this study is strictly conceptual and relies on existing research, the primary limitation is the lack of empirical testing. Consequently, the assumptions made regarding user engagement and behavioural shifts rely heavily on the accuracy and applicability of existing literature, which may contain contextual bias. The VR models will operate under multiple emissions scenarios with the probabilistic projections always being varied due to uncertainty ranges which will obscure the inherent variability within scientific forecasts (IPCC, 2023). As a result, misinterpretations by users are to be established in certain scenarios.

5.2 Physiological and Hardware Limitations

Although hardware development has been excluded from the project scope, the physiological limitations of existing commercial Head-Mounted Displays directly affect the communication framework. Cybersickness, manifesting as nausea, dizziness or ocular strain, remains a significant barrier to prolonged VR exposure. Additionally, the rendering of complex, dynamic environmental simulations requires consequent computing power to maintain immersion and prevent latency. This potentially restricts the accessibility of the proposed system in standard office environments lacking high-end graphic processing capabilities.

Psychological Constraints

A critical behavioural limitation is the “Novelty Effect”, where user engagement and pro-environmental behaviour are short-lived and driven by the excitement of the new technology rather than leading to a fundamental shift in values. As users become accustomed to the virtual environment, the emotional impact of the simulations may diminish, leading to “Empathy Decay”. Therefore, designing a communication framework that sustains long-term behavioural change, rather than a temporary spike in awareness, remains a complex psychological challenge.

6 Conclusion and Future scope

6.1 Conclusion

The persistent disconnect between global climate awareness and tangible pro-environmental action highlights a fundamental limitation in traditional corporate communication strategies. As established in this research, relying on the "Information Deficit Model" and static data representations often exacerbates "Climate Perception Inertia," framing the climate crisis as a psychologically distant threat. To counter this, this study has proposed a conceptual Virtual Reality (VR) framework tailored specifically to transform organizational and workplace energy conservation.

By conceptualizing a system that utilizes a probabilistic backend, this framework actively bridges the gap between abstract energy metrics and observable environmental consequences. The integration of spatial presence and experiential learning within VR reduces psychological distance, immersing the user in an environment where their direct choices either trigger sudden "Flash Point" catastrophes or accelerate "Boiling Frog" environmental degradation. Ultimately, this conceptual model establishes a rigorous academic and technical foundation for shifting corporate sustainability efforts from passive knowledge absorption to active, empathy-driven behavioural change.

6.2 Future Scope

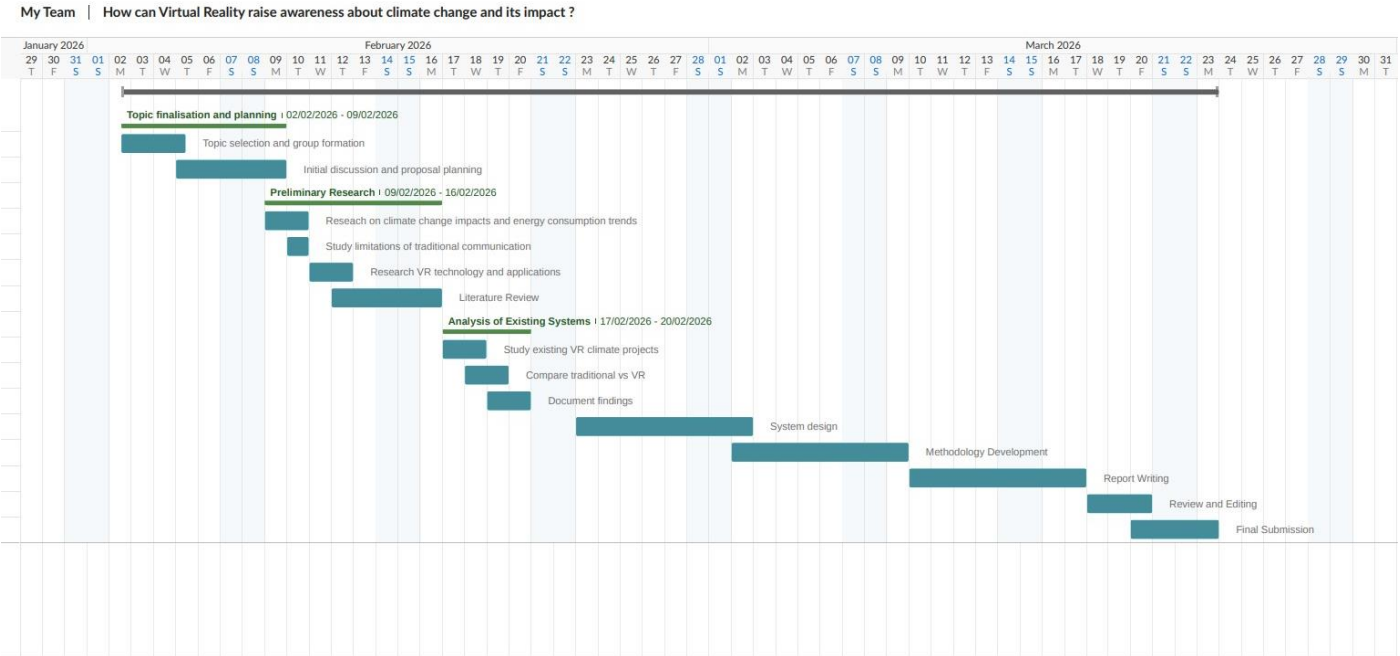
While this research establishes a robust conceptual architecture, achieving sustained behavioural change requires transitioning from theoretical modelling to active corporate integration. The future trajectory of this study encompasses the following strategic and technological expansions:

- **Interactive Prototypes and Longitudinal Tracking**: The immediate next phase requires developing the UI/UX wireframes into a fully functioning prototype utilizing a modern rendering engine (e.g., Unity or WebXR). Following deployment, longitudinal behavioural studies including spanning six to twelve months in active

commercial offices must be conducted to track the persistence of pro-environmental behaviours and evaluate the system's resilience against the "Novelty Effect" and subsequent "Empathy Decay."

- **IoT Integration and Real-Time Feedback Loops**: Future iterations will evolve from probabilistic simulations to live-data ecosystems. By integrating the VR framework with localized Internet of Things (IoT) infrastructure, such as smart meters and HVAC sensors, the system will visualize a 1:1 ratio of real-time office energy waste to projected environmental impacts. This transforms the VR experience from an educational simulation into a live, interactive corporate dashboard.
- **Collaborative Multi-User Environments:** Workplace energy consumption is a collective issue. Future research should explore Multi-User Virtual Environments (MUVEs), allowing entire departments or project teams to enter the simulation simultaneously. Experiencing the environmental consequences of shared energy habits as a group will shift the psychological focus from individual guilt to collaborative problem-solving, directly addressing corporate culture.
- **Integration with Corporate Social Responsibility (CSR) Metrics:** Ultimately, this VR communication framework must bridge the gap between IT solutions and overarching business strategy. Future developments will explore how the behavioural data and energy savings generated by these VR interventions can be quantified and seamlessly integrated into a company's formal Environmental, Social, and Governance (ESG) reporting. By explicitly connecting technical innovation with CSR metrics, this framework will decisively prove the commercial return on investment (ROI) of immersive sustainability communication.

7 Project Schedule



8 Task Distribution Table

Team Member	Role	Responsible Sections
Doongoor Khushi	Climate Change Research Lead	1.1, 1.3, 1.4, 2.1, 6, 7, 8
Jummaheer Teesha Neeharika	Energy and Sustainability Analyst	1.2, 2.2, 3.3.1, 3.3.2

Mohonee Yashmeeta	VR and Technical Systems Lead	2.4.1, 2.4.2, 3.1
Subratty Ghavish	Communication and Behavioural Research Lead	2.3, 2.4.3, 3.2,
Govind Dhruv	System Design and Methodology Lead	3.3.3, 4, 5

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